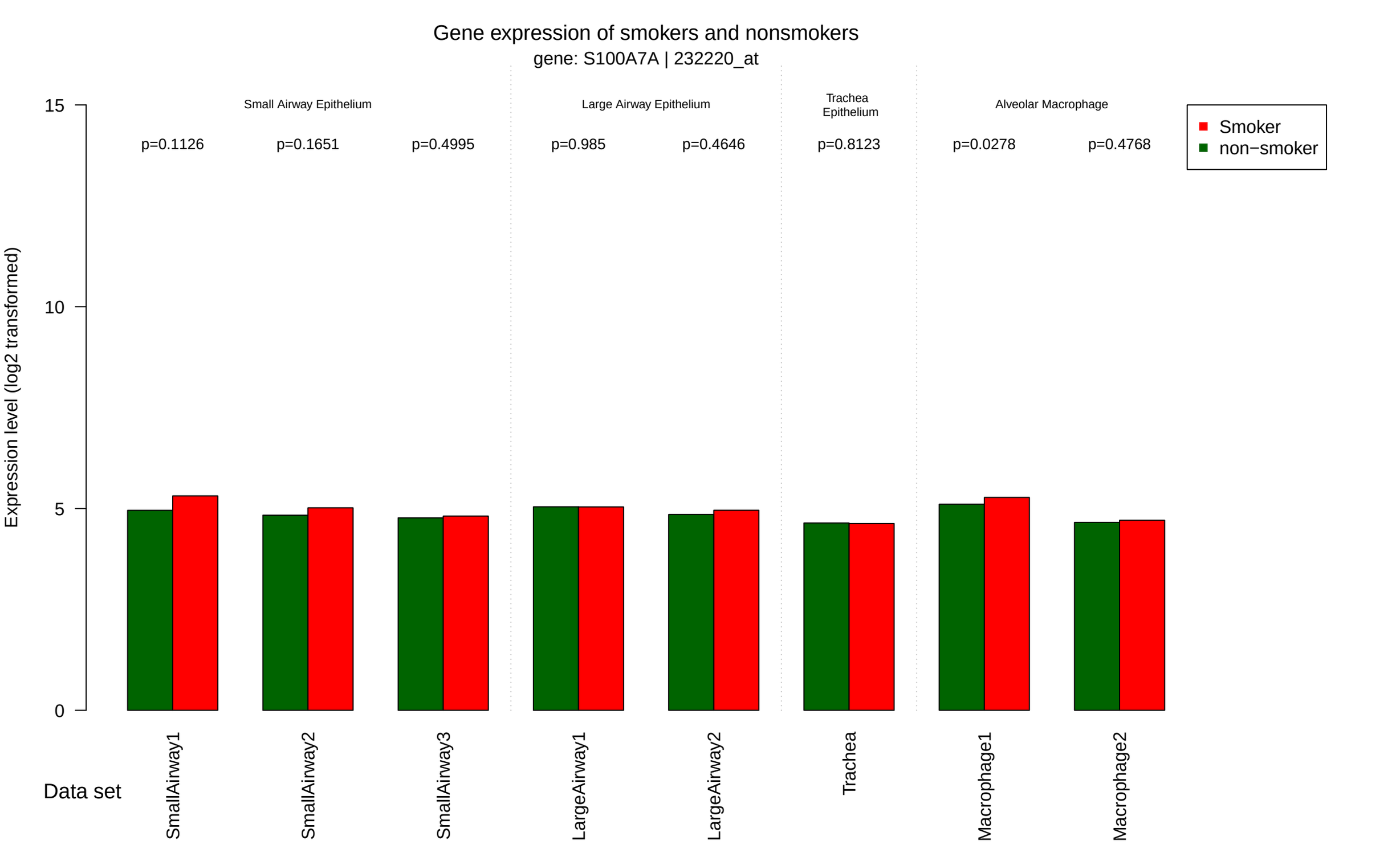
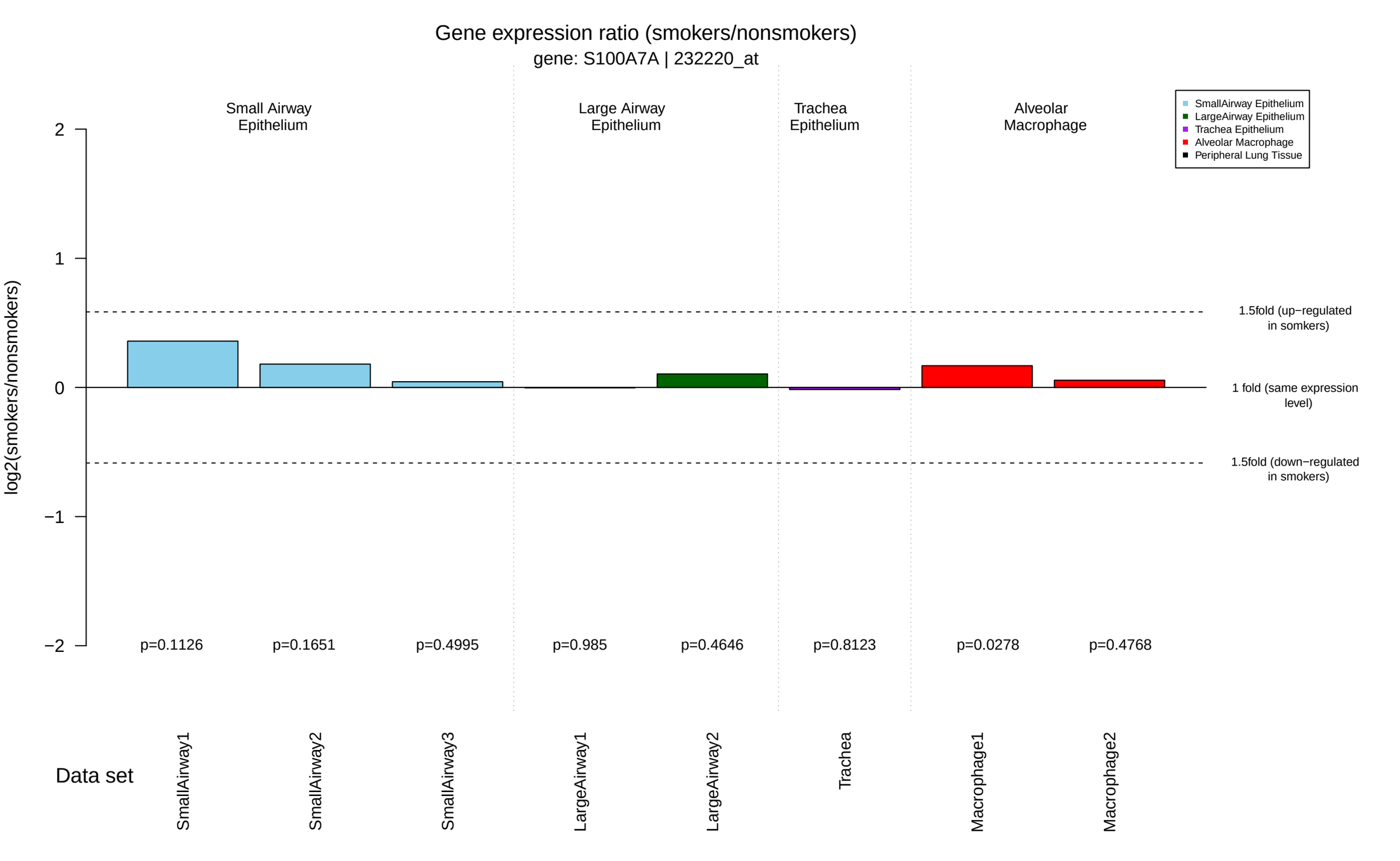
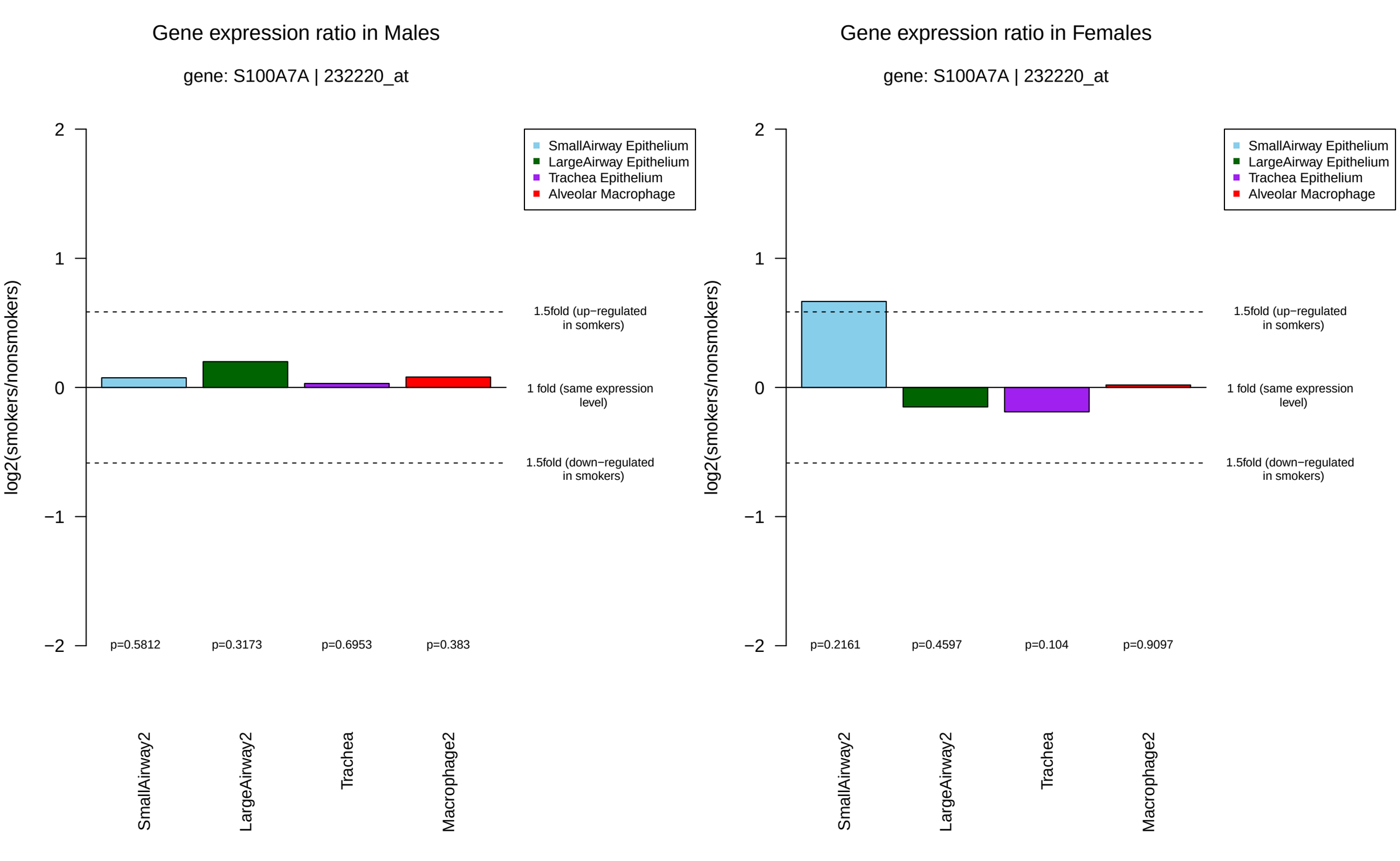
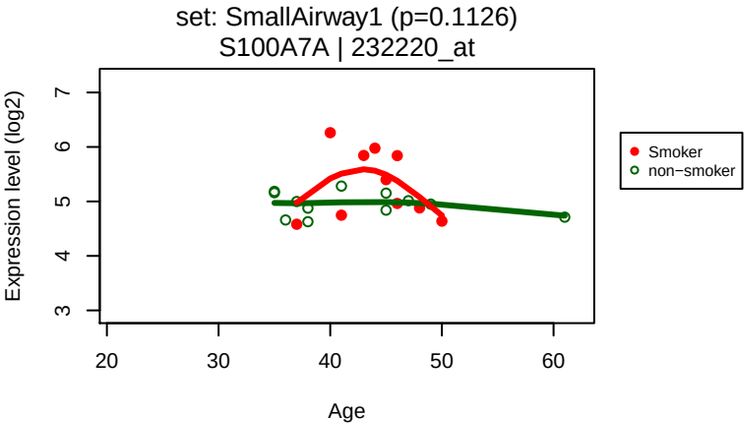
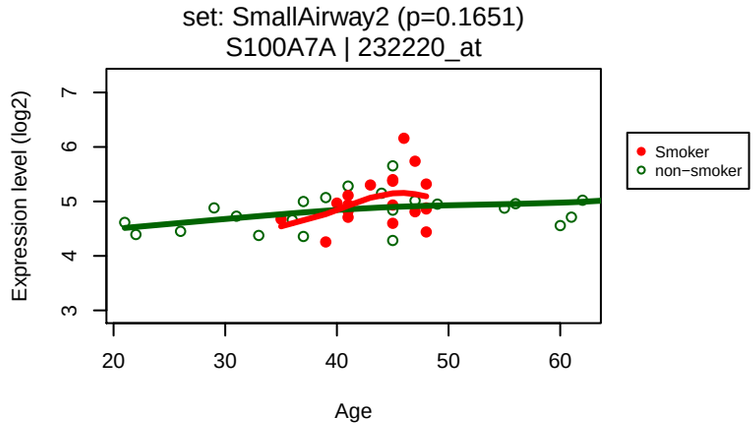
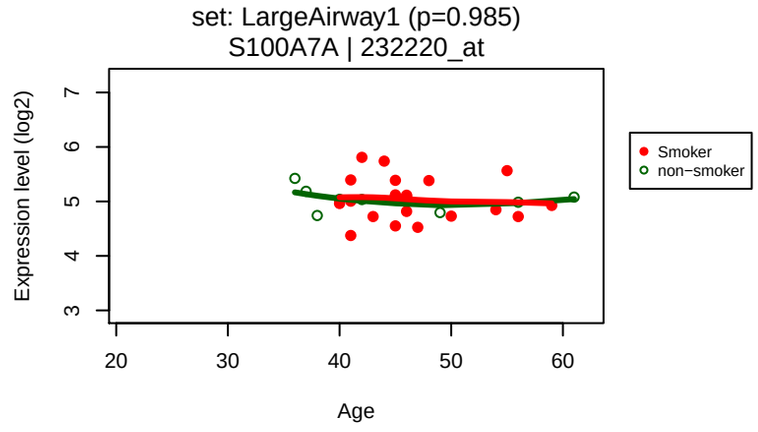

Gene expression of smokers and nonsmokers
gene: S100A7A | 232220_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1126
p=0.1651
p=0.4995
p=0.985
p=0.4646
p=0.8123
p=0.0278
p=0.4768
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

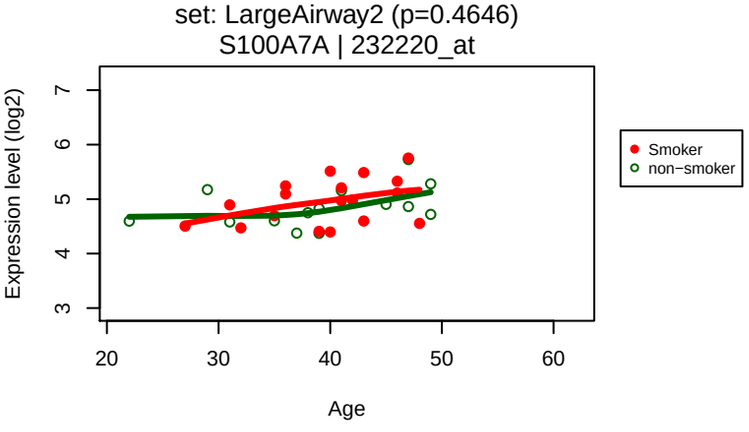

Gene expression ratio (smokers/nonsmokers)
gene: S100A7A | 232220_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1126
p=0.1651
p=0.4995
p=0.985
p=0.4646
p=0.8123
p=0.0278
p=0.4768
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

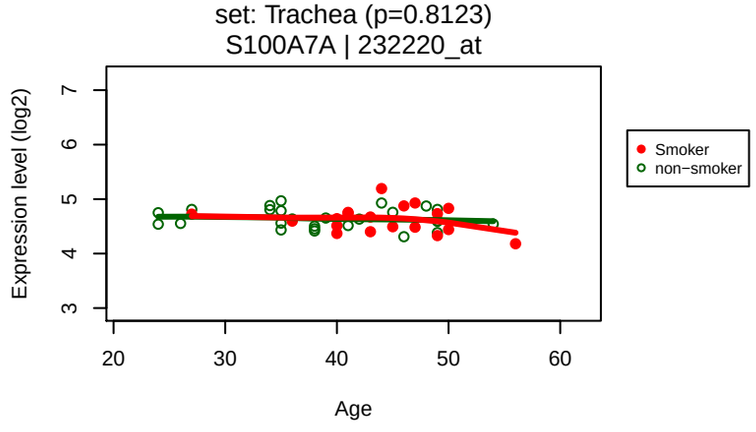

Gene expression ratio in Males
gene: S100A7A | 232220_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5812
p=0.3173
p=0.6953
p=0.383
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: S100A7A | 232220_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2161
p=0.4597
p=0.104
p=0.9097
SmallAirway2
LargeAirway2
Trachea
Macrophage2

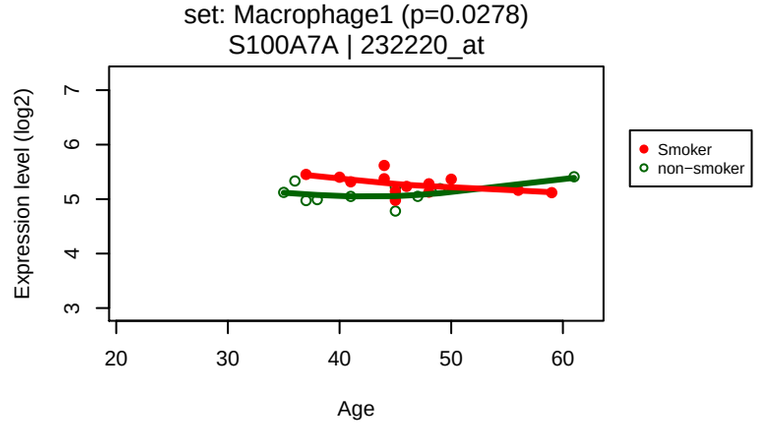

set: SmallAirway1 (p=0.1126)
S100A7A | 232220_at
Expression level (log2)
Age
Smoker
non−smoker

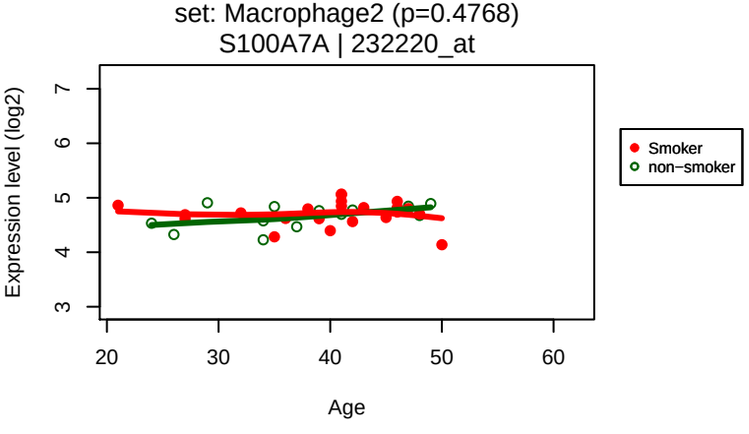

set: SmallAirway2 (p=0.1651)
S100A7A | 232220_at
Expression level (log2)
Age
Smoker
non−smoker

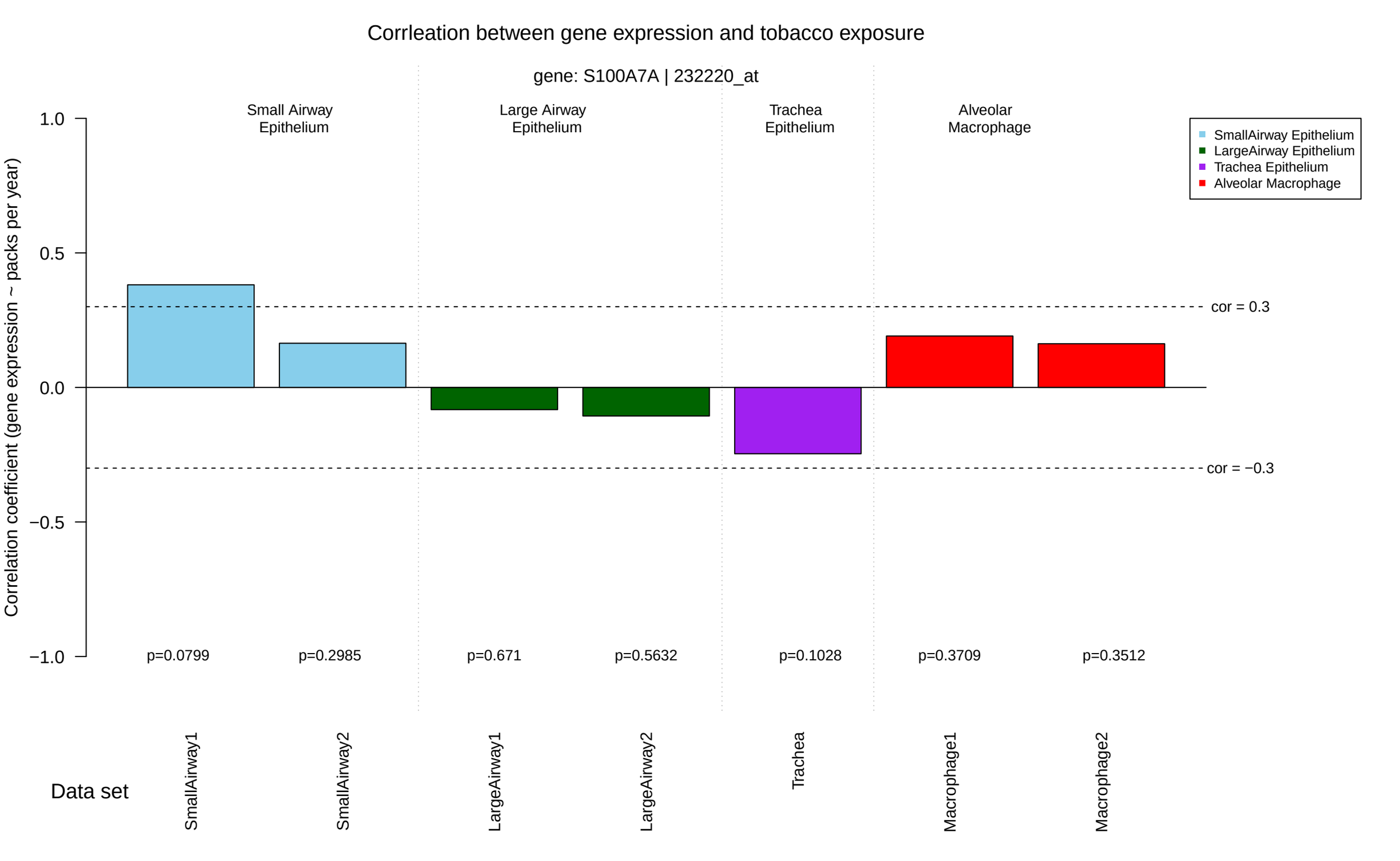

set: LargeAirway1 (p=0.985)
S100A7A | 232220_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4646)
S100A7A | 232220_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8123)
S100A7A | 232220_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0278)
S100A7A | 232220_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4768)
S100A7A | 232220_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: S100A7A | 232220_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0799
p=0.2985
p=0.671
p=0.5632
p=0.1028
p=0.3709
p=0.3512
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2